

# Satellite-Based Urban Growth and Economic Activity Intelligence System

Review 2 — Project Documentation

**BCSE497J Project I**

School of Computer Science and Engineering

Fall Semester 2026–27

Study area — Varanasi, Uttar Pradesh, India

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Guide — [Guide Name]

# Review 3 — Implementation and Results

**Satellite-Based Urban Growth and Economic Activity Intelligence System** BCSE497J Project I · School of Computer Science and Engineering · Fall 2026–27 Panel review · 20 marks · 16 September 2026 · Study area: Varanasi, Uttar Pradesh

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This document is the **initial Chapter 4 (Implementation and Results)** that Review 3 asks for, together with the updates it makes to Chapters 1–3 (§3–§4) and the record of corrections and review comments (§6–§7). Abbreviations and official dataset identifiers are listed in `CONVENTIONS.md`. Every number below is produced by `run_review3.bat` and stored in `outputs/review3_results.json`; every figure is in `figures/`.

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## 1. Summary

Since Review 2 the project has done three things.

**It fixed what was wrong.** An audit of our own code found fifteen defects (§6). The most serious: the pipeline treated GHSL's 2025 epoch as the current year. That epoch is a *projection* made by the GHSL model, so every "2010–2025" change figure shown at Review 2 was measured against a forecast. Every number in this document is recomputed on the observed epochs 2010, 2015 and 2020, and the configuration now refuses a projected epoch used as a measurement.

**It tested the results against independent data.** Six built-up datasets, a second land-surface-temperature sensor, a hold-out test of the ghost-growth classes and — new for this review — Indian data: the Census of India 2001 and 2011, the 2013 Economic Census, and district GDP published by the Government of Uttar Pradesh.

**It finished the prediction module.** Logistic regression and random forest are trained on 2010–2015, scored on 2015–2020, compared on a TOC curve, and the better model projects growth 5 and 10 years ahead of 2020.

| Result                                              | Value                                                                                         |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Built-up surface, 2010 → 2020                       | 72.48 → 89.73 km <sup>2</sup> , <b>+17.25 km<sup>2</sup> (+23.8%)</b>                         |
| Urban extent (cells at least 20% built)             | 128.43 → 154.39 km <sup>2</sup> , <b>1.86% per year</b>                                       |
| Form of the 10.66 km <sup>2</sup> of new urban land | infill 8.3% · edge expansion 47.5% · <b>leapfrog 44.3%</b>                                    |
| Built-up growth compared with population growth     | <b>1.3× to 2.3× faster</b> , depending on the population dataset                              |
| Low-activity new development that is filling up     | <b>74%</b> (Review 2 said 95%); confirmed by a hold-out test, p = 0.010                       |
| Ghost growth                                        | <b>1.44 km<sup>2</sup></b> in 23 zones; 82% of the flagged cells contain buildings            |
| Surface heat, March–May 2024                        | urban cells <b>1.66 °C cooler</b> than rural land on average; 94% of hotspot area is rural    |
| Growth model on the held-out period 2015–2020       | random forest Figure of Merit <b>0.103 (18.7× random)</b> ; logistic regression 0.069 (12.5×) |
| Predicted new urban land (a prediction)             | <b>+14.9 km<sup>2</sup> by 2025, +31.2 km<sup>2</sup> by 2030</b>                             |
| Night-time light vs district GDP, 71 UP districts   | Spearman ρ <b>0.85</b> , R <sup>2</sup> 0.75 (log-log)                                        |
| Night-time light vs jobs, 10,143 towns and villages | Spearman ρ <b>0.41</b> , no better than population                                            |

Three conclusions changed because of the checks, and we present them as such:

1. The split between *emerging* and *ghost growth* holds up, but it is weaker than Review 2 claimed.
  2. Varanasi has **no daytime surface heat island** in the pre-monsoon season; its hottest surfaces are bare farmland.
  3. Night-time light is a good proxy for economic activity at district scale and a weak one at village scale, which limits what the ghost-growth screen can claim about any single cell.
-

## 2. Implementation status

| Module<br>(approved scope)                                      | Status                                            | Code                                                                                                                                                                                                | Evidence                                                                          |
|-----------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Data acquisition<br>— open path<br>(GHSL,<br>OpenStreetMap)     | Done                                              | <code>data/ghsl.py</code> , <code>data/osm.py</code> ,<br><code>scripts/prefetch.py</code>                                                                                                          | GHSL tile <code>R6_C26</code> ; 1,999 points of<br>interest; 38,455 road segments |
| Data acquisition<br>— Google Earth<br>Engine (13 layers)        | Done                                              | <code>data/gee.py</code> ,<br><code>scripts/export_review3_layers.py</code>                                                                                                                         | cloud masks and composite-depth<br>checks in code                                 |
| Data acquisition<br>— Indian datasets                           | <b>New</b>                                        | <code>scripts/fetch_indian_data.py</code> ,<br><code>data/shrug.py</code>                                                                                                                           | <code>data/raw/india/MANIFEST.json</code> ,<br>with SHA-256 hashes                |
| Common analysis<br>grid, 100 m<br>analysis / 500 m<br>reporting | Done                                              | <code>aoi.py</code> , <code>analysis/zonal.py</code>                                                                                                                                                | exact nesting, unit-tested                                                        |
| Built-up change<br>and urban form                               | Done,<br>recomputed                               | <code>analysis/builtup.py</code>                                                                                                                                                                    | \$5.1                                                                             |
| Activity index and<br>growth typology                           | Done, rule<br>revised                             | <code>analysis/ghost.py</code> ,<br><code>analysis/nightlights.py</code>                                                                                                                            | \$5.4–\$5.6                                                                       |
| Green cover<br>change                                           | Done,<br>periods<br>matched                       | <code>analysis/vegetation.py</code>                                                                                                                                                                 | \$5.8                                                                             |
| Surface heat<br>island                                          | Done,<br>corrected                                | <code>analysis/thermal.py</code>                                                                                                                                                                    | \$5.7                                                                             |
| Growth<br>prediction, +5<br>and +10 years                       | <b>Extended</b> —<br>random<br>forest, TOC,<br>+5 | <code>analysis/growth_model.py</code> ,<br><code>scripts/run_growth_model.py</code>                                                                                                                 | \$5.9                                                                             |
| Validation suite                                                | <b>New</b>                                        | <code>scripts/validate_typology.py</code> ,<br><code>cross_checks.py</code> ,<br><code>validate_population.py</code> ,<br><code>validate_economy.py</code> ,<br><code>analysis/validation.py</code> | \$5                                                                               |
| Results pack —<br>figures, zone<br>cards, numbers               | <b>New</b>                                        | <code>scripts/make_figures.py</code> ,<br><code>scripts/make_zone_cards.py</code>                                                                                                                   | 14 figures, 23 zone cards,<br><code>review3_results.json</code>                   |
| Dashboard                                                       | Updated                                           | <code>dashboard/app.py</code>                                                                                                                                                                       | 7 tabs incl. Validation; 0 exceptions in<br>a headless test                       |
| City search (any<br>city)                                       | Review 4                                          | —                                                                                                                                                                                                   | route checked, \$9                                                                |

| Module<br>(approved scope)               | Status   | Code | Evidence |
|------------------------------------------|----------|------|----------|
| Ground or image labels for flagged zones | Review 4 | —    | —        |

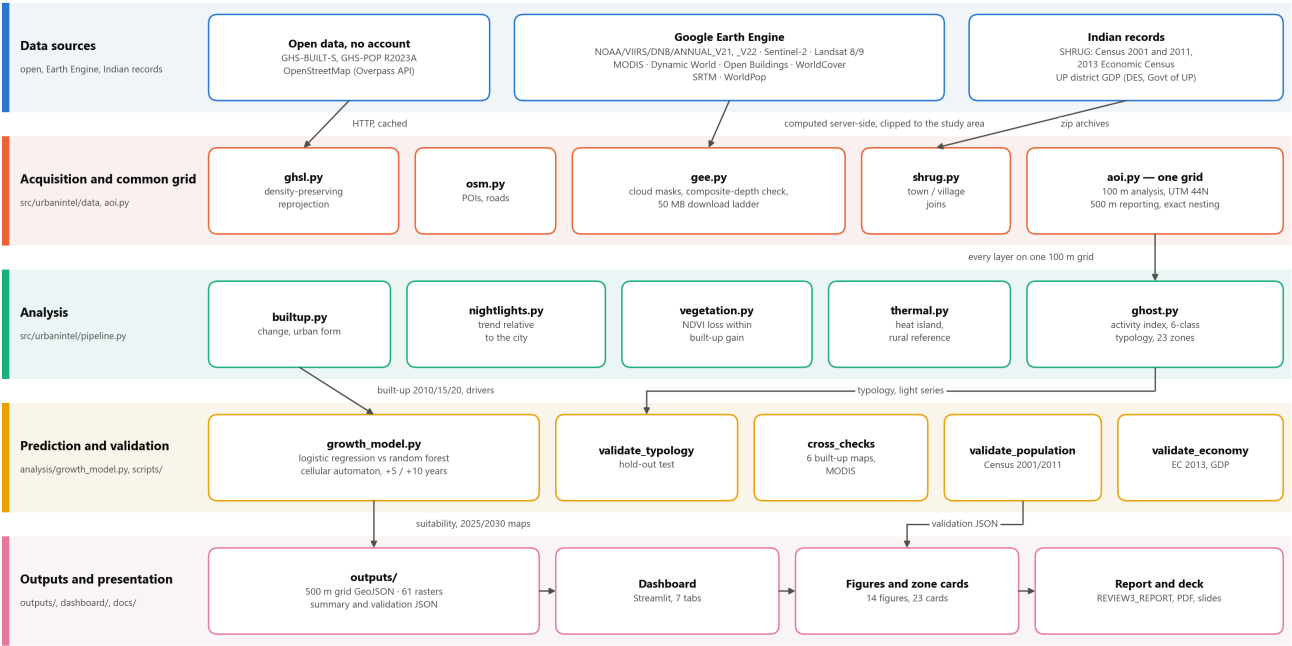
Paths are relative to `src/urbanintel/` unless they start with `scripts/` or `dashboard/`.

**Running it.** `run_review3.bat` runs the whole chain — unit tests, pipeline, growth models, the four validations, figures and zone cards — and prints the start and finish time of every step, so it can be shown live. The unit tests pass **41 of 41**; 16 of them were added for Review 3, each tied to a defect in §6.

**Architecture.** The figure below shows the five layers of the system, each a folder of the code. Nine architecture diagrams (pipeline data flow, the typology algorithm, the relative light trend, the growth model, the heat-island method, the validation framework, the temporal design, the grid) and six progress charts are generated by `scripts/make_diagrams.py` into `diagrams/`; `DIAGRAMS.md` explains each one.

System architecture

Five layers, each a folder of the code. Data moves top to bottom; every arrow is a real hand-off.



## 3. Datasets — update to Chapter 3

### 3.1 Satellite and gridded datasets

| Official dataset                                                                                      | Provider                            | Resolution     | Years used                                                    | What we use it for                               |
|-------------------------------------------------------------------------------------------------------|-------------------------------------|----------------|---------------------------------------------------------------|--------------------------------------------------|
| <code>GHS-BUILT-S R2023A</code> (tile <code>R6_C26</code> )                                           | European Commission JRC             | 100 m          | 2010, 2015, 2020 observed; 2025 only as a labelled projection | built-up change, urban form, growth-model target |
| <code>GHS-POP R2023A</code> / <code>JRC/GHSL/P2023A/GHS_POP</code>                                    | European Commission JRC             | 100 m          | 2001–2020 (interpolated between epochs)                       | activity index; population checks                |
| <code>WorldPop/GP/100m/pop</code>                                                                     | WorldPop, University of Southampton | ~92 m          | 2001, 2010, 2011, 2020                                        | independent population estimate                  |
| <code>NOAA/VIIRS/DNB/ANNUAL_V21</code> + <code>NOAA/VIIRS/DNB/ANNUAL_V22</code>                       | NOAA                                | ~463 m         | 2013–2021 + 2022–2024                                         | night-time light: activity level and trend       |
| <code>COPERNICUS/S2_SR_HARMONIZED</code>                                                              | ESA / Copernicus                    | 10–20 m        | Oct–Mar 2018-19 and 2024-25                                   | NDVI, NDBI, true-colour zone evidence            |
| <code>LANDSAT/LC08/C02/T1_L2</code> + <code>LANDSAT/LC09/C02/T1_L2</code> (band <code>ST_B10</code> ) | USGS                                | 30 m           | Mar–May 2013 and 2024                                         | LST (Land Surface Temperature)                   |
| <code>MODIS/061/MOD11A2</code>                                                                        | NASA LP DAAC                        | 1 km           | Mar–May 2024                                                  | independent LST check                            |
| <code>GOOGLE/DYNAMICWORLD/V1</code>                                                                   | Google                              | 10 m           | 2018, 2024                                                    | water mask, built-up gain, built-up comparison   |
| <code>GOOGLE/Research/open-buildings-temporal/v1</code>                                               | Google                              | ~4 m effective | 2016, 2023                                                    | do flagged cells contain buildings?              |
| <code>ESA/WorldCover/v200</code>                                                                      | ESA                                 | 10 m           | 2021                                                          | built-up comparison                              |
| <code>USGS/SRTMGL1_003</code>                                                                         | NASA / USGS                         | 30 m           | single epoch                                                  | slope driver for the growth model                |
| OpenStreetMap, via the Overpass API                                                                   | OpenStreetMap contributors          | vector         | current snapshot                                              | points of interest, road density                 |

Every Earth Engine composite passes a depth check before it is used: at least 20 distinct acquisition dates for Sentinel-2 and 4 for Landsat. The 2024 LST composite is built from 22 Landsat 8 and 9 scenes on 11 dates; the 2013 one from 10 Landsat 8 scenes on 5 dates (Landsat 9 did not exist yet).

### 3.2 Indian and other non-satellite data (new)

| Dataset                                                                                            | Publisher                                                          | What we use                                                     | Access                                  |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------|
| SHRUG — 2011 Population Census Abstract, by town/village and district                              | Development Data Lab, from the Census of India 2011                | population of 71 UP districts and 11,624 towns and villages     | free download, CC BY-NC-SA 4.0          |
| SHRUG — 2001 Population Census Abstract, by town/village                                           | Development Data Lab, from the Census of India 2001                | 2001–2011 growth over the study area                            | as above                                |
| SHRUG — 2013 Economic Census (6th Economic Census, MoSPI)                                          | Development Data Lab                                               | non-farm employment by town/village and district                | as above                                |
| SHRUG — town/village and 2011 district boundaries                                                  | Development Data Lab                                               | polygons for summing satellite values                           | as above                                |
| District Domestic Product of Uttar Pradesh, base year 2011-12 (2020-21 revised, 2021-22 tentative) | Directorate of Economics & Statistics, Government of Uttar Pradesh | Gross District Domestic Product at current prices, 75 districts | published spreadsheets, updes.up.nic.in |

SHRUG stands for the Socioeconomic High-resolution Rural-Urban Geographic Platform for India. It links Indian census tables to one set of town and village identifiers, which is what lets us compare satellite values with census counts place by place. The files, their sources, sizes and SHA-256 hashes are recorded in `data/raw/india/MANIFEST.json`.

Considered and kept for Review 4: NRSC Bhuvan NUIS/AMRUT land use, the VDA Master Plan 2031, the UP RERA project registry, ward boundaries, IMD gridded temperature, Bhoonidhi CartoDEM and LISS-III imagery, and MapmyIndia (a paid service). Property portals will not be scraped.

### 3.3 Data-quality safeguards now in the code

| Safeguard                                                                               | Why                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>Config.check_epochs()</code> rejects a projected GHSL epoch used as a measurement | the Review 2 error (§6, D1) cannot recur silently                                                                                                                                                  |
| Composite depth: Sentinel-2 $\geq 20$ dates, Landsat $\geq 4$ dates                     | a one- or two-day composite carries that day's weather, not the season                                                                                                                             |
| Landsat cloud mask uses QA_PIXEL bits 1–4 (diluted cloud, cirrus, cloud, shadow)        | bits 3–4 alone let cloud edges and thin cirrus through                                                                                                                                             |
| Class masks are read with no-data disabled                                              | Earth Engine tags 0 as no-data on some integer exports; read naively, WorldCover's built-up area came out as 480 km <sup>2</sup> instead of 191 km <sup>2</sup> — caught by the cross-check itself |
| Population is summed on each dataset's own grid                                         | sampling WorldPop's per-pixel counts on a 100 m grid understates totals by about a fifth — our own planning audit made this mistake (§6, D15)                                                      |

## 4. Method changes since Review 2 — update to Chapter 3

### 4.1 Observed epochs only (D1)

**What changed.** Baseline 2010, current 2020. Review 2 used 2025 as current.

**Why.** `GHS-BUILT-S R2023A` epochs after 2020 are the GHSL model's own projections, not satellite observations.

**How.** `config/varanasi.yaml` lists the observed epochs; 2025 is loaded under layer names ending in `_projected` and appears only in a labelled comparison. `check_epochs()` runs at the start of every pipeline run and is unit-tested.

**Result.** Every built-up, urban-form, typology and growth-model figure is recomputed (§5, §6).

### 4.2 Activity level, and what "activity rising" means (D4)

**What changed.** The activity index uses mean night-time radiance over 2022–2024 (Review 2: 2024 alone), set against development observed up to 2020 — so new building has had at least two years to be occupied. A new, low-activity cell is now *emerging* only if its night-time light rose **significantly faster than the established city**; otherwise it is *ghost growth*.

**Why.** Between 2013 and 2024 much of the city brightened: 51% of urban cells show a statistically significant rise. Review 2's rule — "slope above zero" — therefore called almost every candidate *emerging*. The series also changes product version, from `ANNUAL_V21` to `ANNUAL_V22`, in 2022.

**How.** Each year, the median log radiance of the cells that were already urban in 2010 is subtracted from every cell's log radiance, and a straight-line trend is fitted to what remains ( `nightlights.relative_trend` ). A citywide rise, or a version change that shifts the whole scene, cancels out. The log is taken of radiance + 1, so proportional changes count the same for dim and bright places. "Rising" means a positive relative slope with  $p \leq 0.10$ . All four rules are reported side by side (§5.4).

**Result.** §5.4, validated by a hold-out test in §5.5.

### 4.3 Surface heat island (D2, D3, D10)

**What changed.** The rural reference is now cells under 2% GHSL built surface, minus water (Dynamic World water probability  $\geq 0.5$ ; 2,006 cells) and minus land Dynamic World already sees as built in 2024 (built probability  $\geq 0.2$ ; 12,512 cells). The mean intensity is taken over urban cells (at least 20% built). The Landsat mask uses QA\_PIXEL bits 1–4 and a composite-depth check. LST 2013 is added so change is measured on intensity, not raw temperature.

**Why.** The code documented a water exclusion it never applied, so the Ganga sat in the rural baseline. The figure labelled "mean urban intensity" was in fact the mean over every cell warmer than the rural reference — positive by construction. Raw LST depends on the day's weather, while the urban-minus-rural difference within one scene largely does not.

**Result.** §5.7.

### 4.4 Green cover lost to built-up (D11)

**What changed.** NDVI loss 2018–2024 is intersected with built-up gain over the same years from Dynamic World (built probability rose by at least 0.15 and ended at 0.30 or more). Review 2 intersected it with GHSL

gain over 2010–2025.

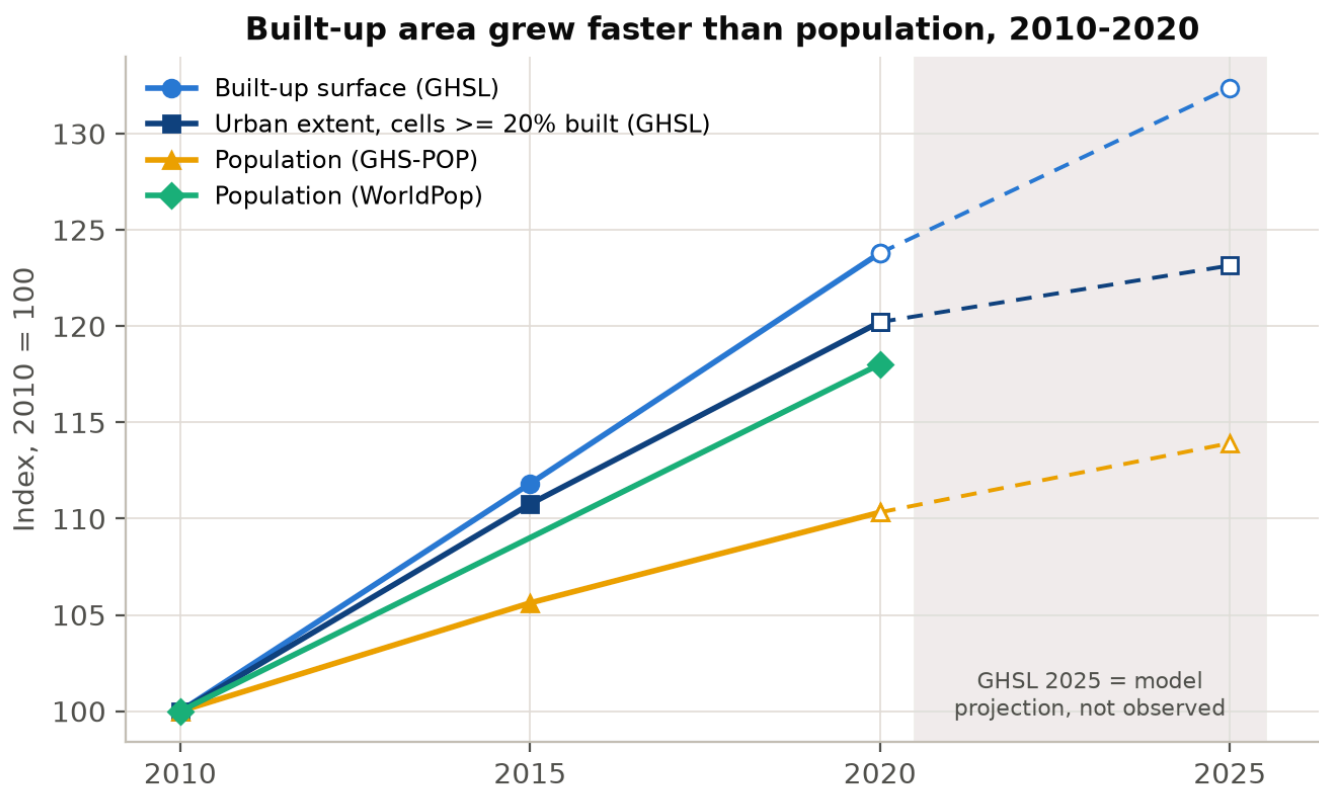
**Result.** 0.67 km<sup>2</sup> of green cover lost to built-up (10.5% of all green loss), against 0.05 km<sup>2</sup> at Review 2 (\$5.8).

#### 4.5 Growth model (D5–D9)

- The cellular automaton now places exactly the number of cells demanded. It used to drop `demand mod 8` cells (6 of 1,214 at Review 2).
- The AUC (Area Under the ROC Curve) on the held-out period is reported next to the training AUC.
- Terrain slope from `USGS/SRTMGL1_003` is added as a driver, and logistic regression is also refitted without roads, because the OpenStreetMap road network is a present-day snapshot.
- A **random forest** — many decision trees whose votes are combined — is trained on exactly the same data: 300 trees, at least 20 cells per leaf, balanced class weights. PLUS (Liang et al. 2021), the model family our design follows, uses a random forest for this step because it can capture thresholds and interactions that a straight-line model cannot.
- TOC (Total Operating Characteristic) curves compare logistic regression, random forest and random allocation on the held-out period.
- Projections are made +5 and +10 years from 2020, in two five-year steps, so growth placed by 2025 attracts growth by 2030.

## 5. Results obtained so far

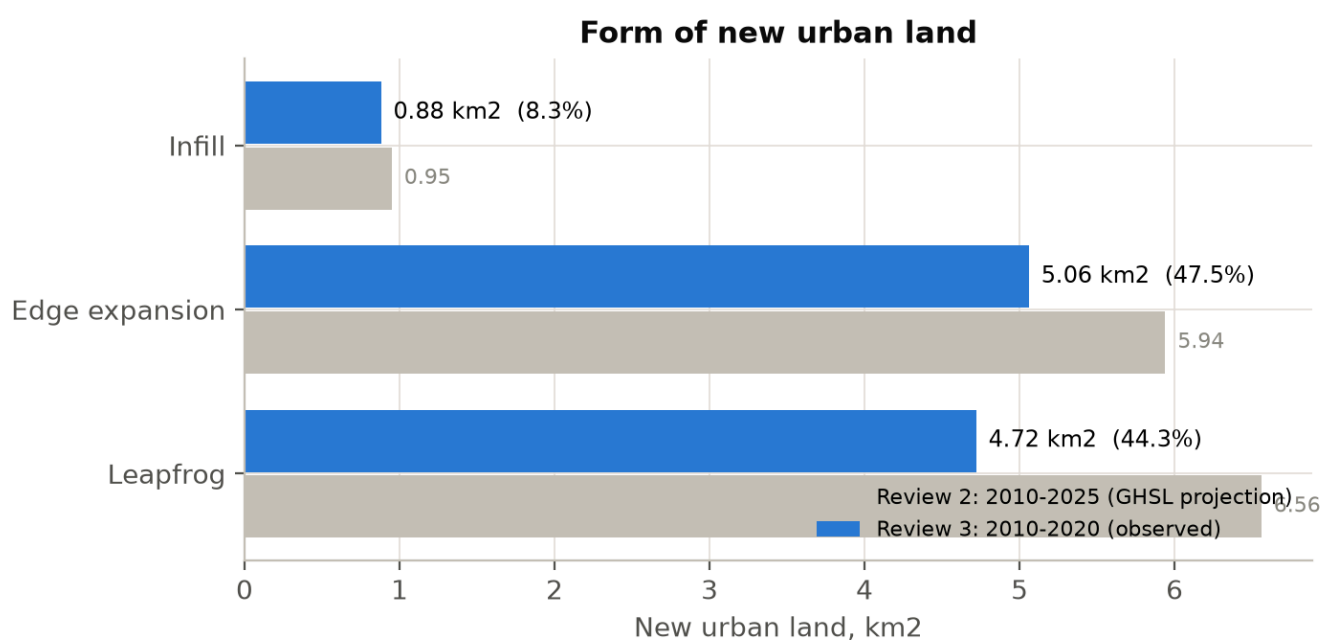
### 5.1 Urban growth, 2010–2020



Sources: GHS-BUILT-S and GHS-POP R2023A (JRC), WorldPop/GP/100m/pop. Study area 1,206 km<sup>2</sup>.

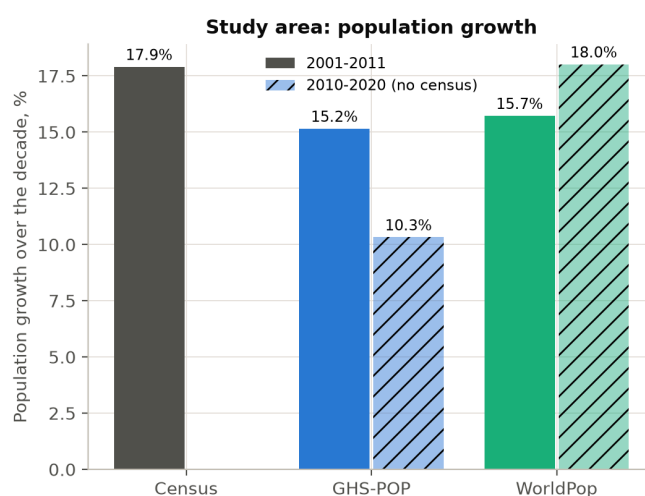
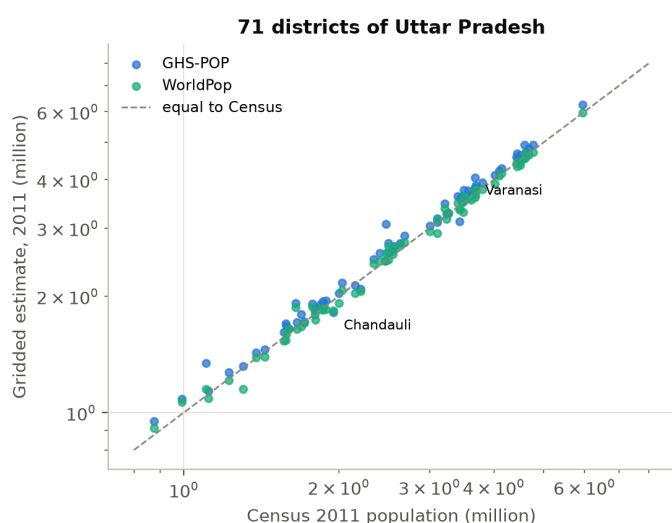
|                                   | 2010      | 2015      | 2020      | 2025 (GHSL projection) |
|-----------------------------------|-----------|-----------|-----------|------------------------|
| Built-up surface, km <sup>2</sup> | 72.48     | 81.05     | 89.73     | 95.95                  |
| Urban extent, km <sup>2</sup>     | 128.43    | 142.25    | 154.39    | 158.16                 |
| Population, GHS-POP               | 4,171,817 | 4,406,438 | 4,602,732 | 4,752,628              |

Built-up surface grew 23.8% and urban extent 20.2% (1.86% a year) over the decade. The GHSL projection implies growth slowing to +6.9% in 2020–2025; our own model projects faster growth (§5.9).



Of the 10.66 km<sup>2</sup> that became urban, 0.88 km<sup>2</sup> is infill (8.3%), 5.06 km<sup>2</sup> edge expansion (47.5%) and **4.72 km<sup>2</sup> leapfrog (44.3%)** — development detached from the existing city. Close to half of all new urban land is scattered, which is the costliest form to supply with roads, water and drainage. Review 2's 48.8% was measured against the projection.

## 5.2 Population, and how fast land is consumed



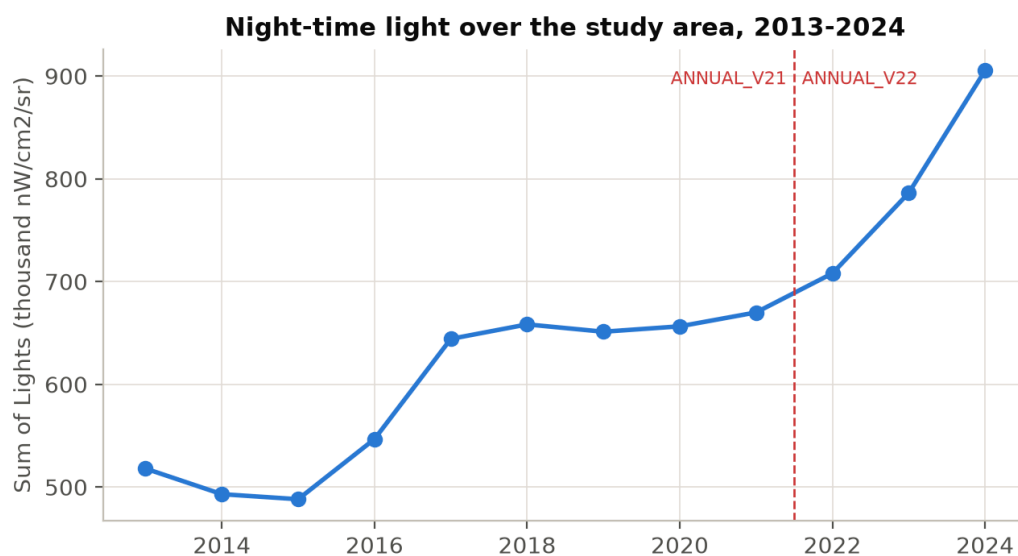
**District totals, Census 2011, 71 districts of Uttar Pradesh.** GHS-POP has a median error of +3.4% (mean absolute 4.8%; 67 districts within 10%). WorldPop has −0.4% (2.6%; 69 within 10%). Both follow the Census closely across districts.

| Area                                                | Census 2011 | GHS-POP       | WorldPop |
|-----------------------------------------------------|-------------|---------------|----------|
| Varanasi district                                   | 3,676,841   | +10.2%        | +2.1%    |
| Chandauli district                                  | 1,952,756   | −7.0%         | −6.1%    |
| Varanasi Municipal Corporation (2011 town boundary) | 1,198,491   | <b>+62.8%</b> | +2.2%    |

GHS-POP spreads population in proportion to built-up volume, so it places far too many people in the dense old city.

**Growth.** Over the 1,011 towns and villages of the study area, the Census records +17.9% from 2001 to 2011; GHS-POP gives +15.2% and WorldPop +15.7% for the same places and years — both slightly low, both close. For 2010–2020 there is no census (the 2021 Census has not been held): GHS-POP gives +10.3% and WorldPop +18.0%. With built surface up 23.8%, built-up area grew **2.3× faster than population with GHS-POP and 1.3× with WorldPop**. WorldPop's rate is in line with the last census decade; GHS-POP's implies a sharp slowdown. We therefore report the range. Land is being consumed faster than population grows under either dataset; the magnitude is uncertain.

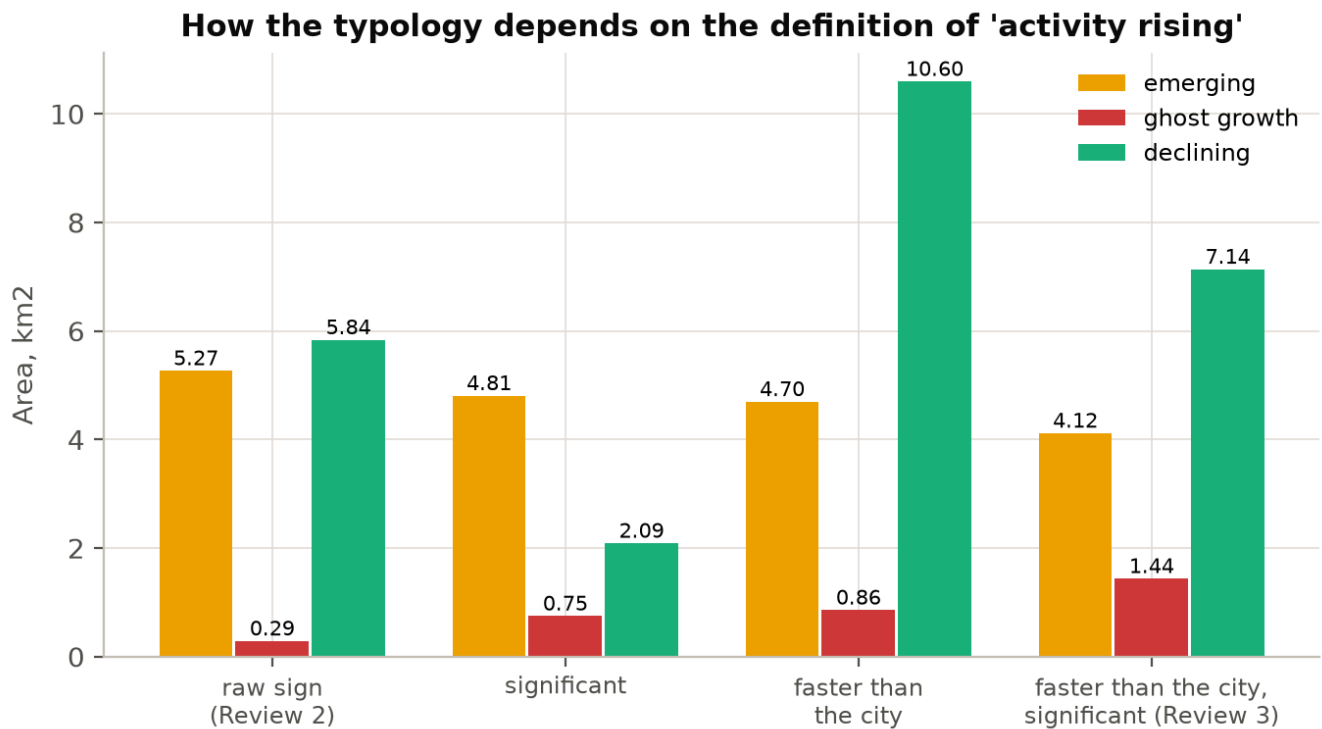
### 5.3 Night-time light



NOAA/VIIRS/DNB/ANNUAL\_V21 (2013-2021) and ANNUAL\_V22 (2022-2024), band average\_masked. A proxy for activity, not a measurement of it.

Sum of Lights over the study area rose from 518,008 (2013) to 905,354 (2024), +74.8%, and 51.4% of urban cells show a statistically significant rise ( $p \leq 0.05$ ). The largest yearly jumps are 2016→2017 (+17.8%) and 2023→2024 (+15.1%). The product change between 2021 and 2022 is marked on the chart; the relative trend of \$4.2 is insensitive to any such scene-wide step.

## 5.4 Growth typology and ghost growth



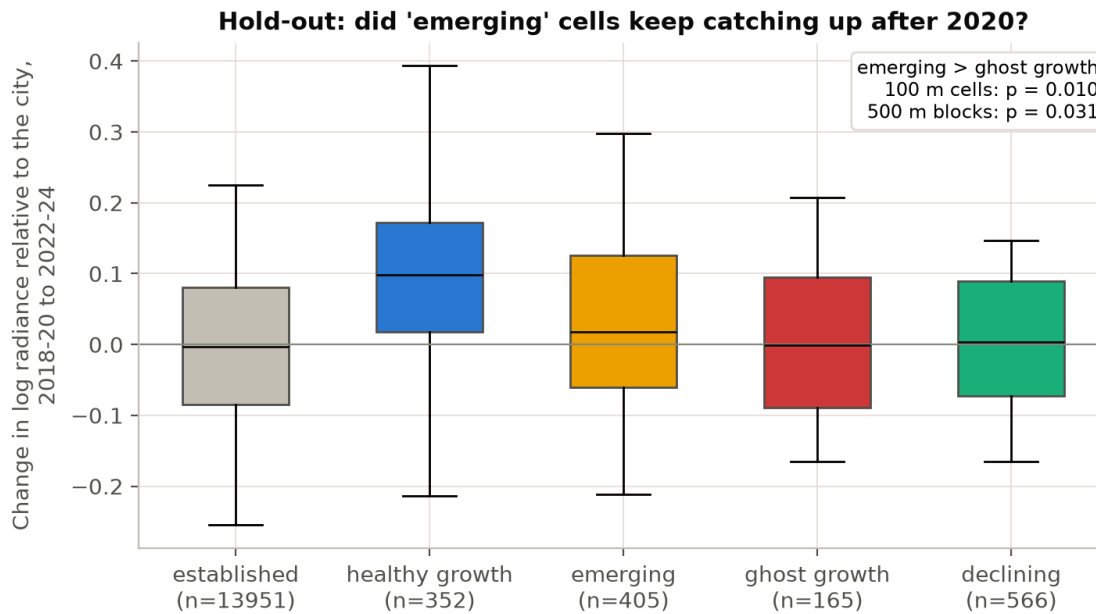
| Class, km <sup>2</sup>         | Review 2 (2025, raw sign) | raw sign | significant | faster than city | faster and significant (Review 3) |
|--------------------------------|---------------------------|----------|-------------|------------------|-----------------------------------|
| emerging                       | 6.88                      | 5.27     | 4.81        | 4.70             | <b>4.12</b>                       |
| ghost growth                   | 0.35                      | 0.29     | 0.75        | 0.86             | <b>1.44</b>                       |
| declining                      | 5.70                      | 5.84     | 2.09        | 10.60            | <b>7.14</b>                       |
| share of candidates filling up | 95%                       | 95%      | 87%         | 85%              | <b>74%</b>                        |

Under the Review 3 rule the other classes are: established 138.03 km<sup>2</sup>, healthy growth 3.66 km<sup>2</sup>, undeveloped 1,104.86 km<sup>2</sup>.

The candidates — new development whose activity is in the bottom quartile for its built-up intensity — are the same 5.56 km<sup>2</sup> under every rule; the rule only decides how they split. Asking for evidence of catching up moves a quarter of them to ghost growth. *Declining* swings between 2.1 and 10.6 km<sup>2</sup> with the rule, so it is treated as indicative only.

Flagged cells group into **23 zones** where they occur at least twice as often as across the city's developed land. The largest are zone 1 (25.288° N, 83.058° E; 4.76 km<sup>2</sup>, 0.25 km<sup>2</sup> flagged), zone 2 (25.311° N, 83.103° E; 3.38 km<sup>2</sup>, 0.18 km<sup>2</sup>) and zone 3 (25.452° N, 83.098° E; 1.48 km<sup>2</sup>, 0.09 km<sup>2</sup>). All are peripheral.

## 5.5 Hold-out test of the typology



Classified with 2013–2020 night lights only (ANNUAL\_V21); outcome never seen by the classifier. Boxes: quartiles; whiskers: 5th–95th percentile.

There are no ground labels for Varanasi, so the split was tested in time. The classification was re-run using only the 2013–2020 night lights (all **ANNUAL\_V21**), then checked against what happened next, which the classifier never saw: the change in brightness relative to the city from 2018–20 to 2022–24.

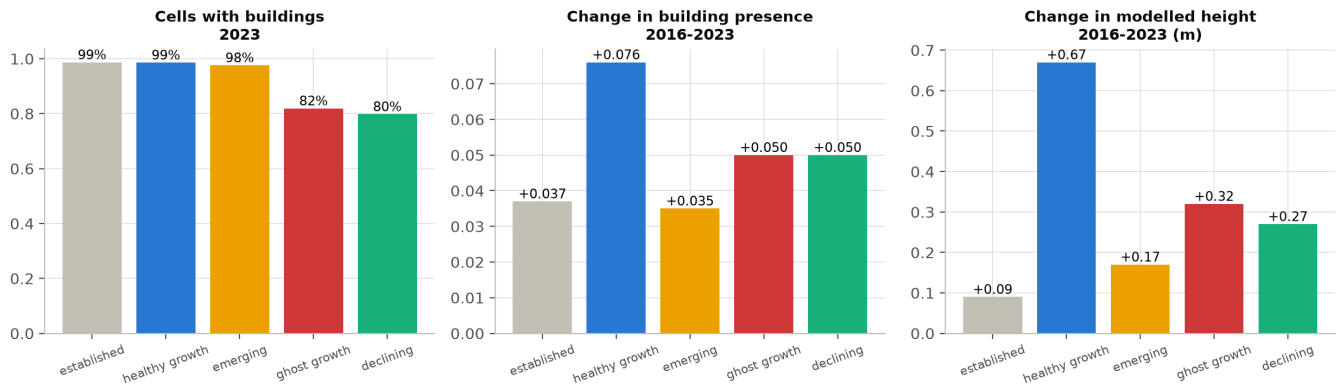
| Class (hold-out classification) | Area, km <sup>2</sup> | Median later change |
|---------------------------------|-----------------------|---------------------|
| healthy growth                  | 3.52                  | +0.099              |
| emerging                        | 4.05                  | +0.018              |
| ghost growth                    | 1.65                  | −0.001              |
| established                     | 139.51                | −0.003              |
| declining                       | 5.66                  | +0.004              |

Cells called *emerging* kept brightening relative to the city more than cells called *ghost growth*: one-sided Mann-Whitney U test  $p = 0.010$  over 405 and 165 cells, and  $p = 0.031$  over 204 and 81 blocks of 500 m. The block test matters because neighbouring 100 m cells share one VIIRS pixel. An emerging cell out-brightened a ghost cell 56% of the time. The other rules: significant only  $p = 0.002$ , raw sign  $p = 0.022$ , faster-than-city without significance  $p = 0.145$ .

91% of the hold-out *emerging* cells are also *emerging* in the full classification. 71% of the hold-out *ghost* cells stay *ghost*; 24% became *emerging* once 2021–24 was added — they began to fill up late.

**Meaning.** The split carries real information: places classified as filling up went on filling up. The effect is modest, so the ghost map is a screening map, not a verdict.

## 5.6 Do the flagged cells contain buildings?

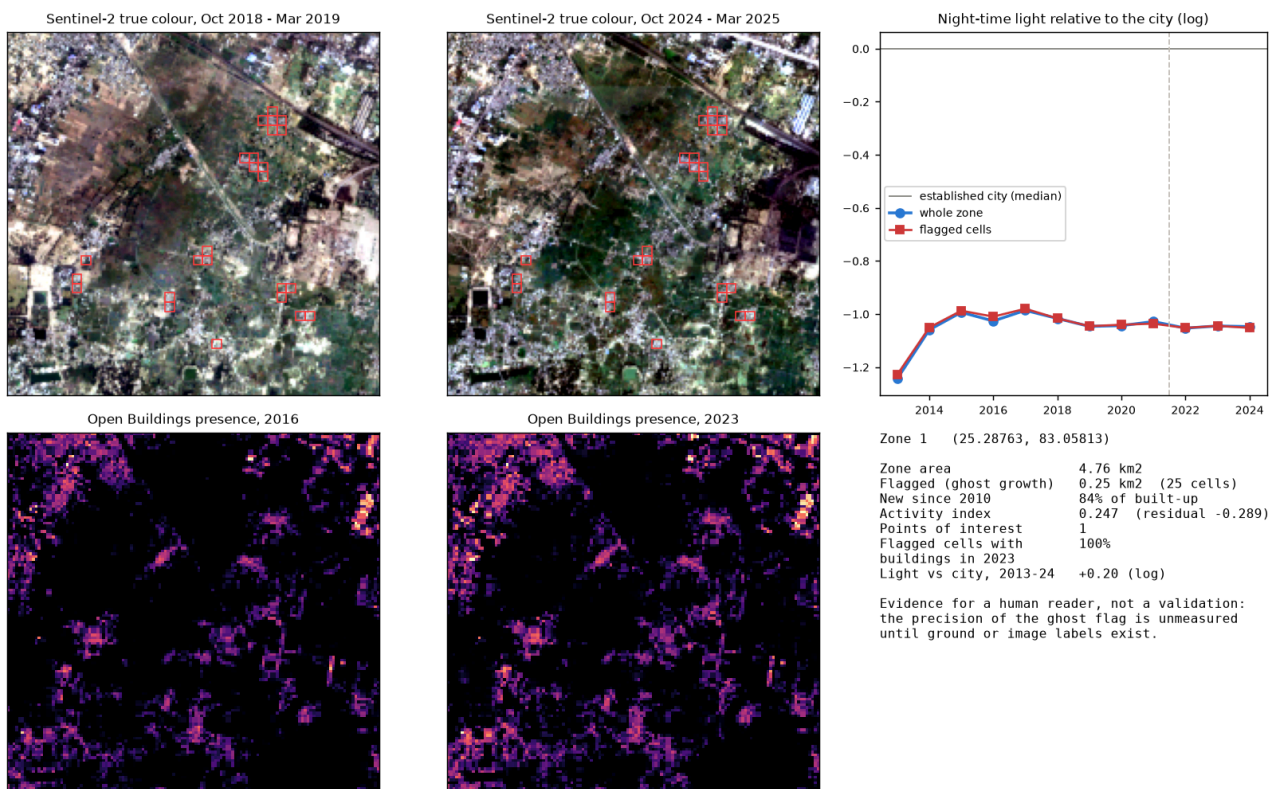


[GOOGLE/Research/open-buildings-temporal/v1](https://ai.google/research/open-buildings-temporal/v1) gives building presence and modelled height for 2016 and 2023.

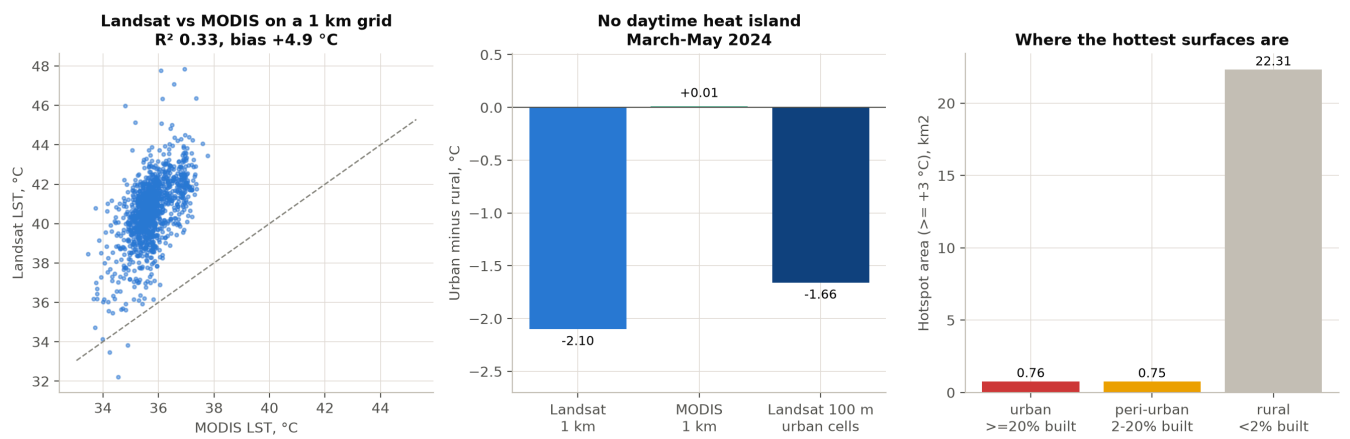
- **82%** of ghost cells contain buildings in 2023 (presence  $\geq 0.05$ ), against 98–99% for the emerging, healthy-growth and established classes.
- **26 of 144 ghost cells (18%) contain no buildings at all** — likely GHSL false positives or land cleared but not yet built. They sit in 6 of the 23 zones (zones 3, 9, 16, 18, 19 and 21; half of zone 20).
- Building presence in ghost cells rose by 0.05 between 2016 and 2023 and modelled height by 0.32 m — as much as in other classes. This fits "built but little used" better than "never built".

**Zone evidence cards.** [figures/zones/zone\\_01.png](#) to [zone\\_23.png](#), one per zone, show Sentinel-2 true colour for 2018-19 and 2024-25 with the flagged cells outlined, Open Buildings presence for 2016 and 2023, the zone's night light against the city, and its numbers. They are evidence for a human reader, not a validation (§8).

### Ghost-growth zone 1 — evidence card



5.7 Surface heat, March–May



|                                 | 2013                  | 2024                  |
|---------------------------------|-----------------------|-----------------------|
| Rural reference                 | 46.34 °C              | 41.42 °C              |
| Mean intensity over urban cells | –1.35 °C              | –1.66 °C              |
| Hotspot area (≥ +3 °C)          | 51.52 km <sup>2</sup> | 23.82 km <sup>2</sup> |

The same 2024 scene under the rural rules, before and after the fix:

| Rural reference rule                               | Reference | Mean over urban cells | Hotspots              |
|----------------------------------------------------|-----------|-----------------------|-----------------------|
| Review 2: under 2% built                           | 41.18 °C  | –1.43 °C              | 28.16 km <sup>2</sup> |
| ... minus water                                    | 41.23 °C  | –1.48 °C              | 27.05 km <sup>2</sup> |
| ... minus water and recently built land (Review 3) | 41.42 °C  | –1.66 °C              | 23.82 km <sup>2</sup> |

**Finding.** In the pre-monsoon daytime, Varanasi's built-up area is on average cooler than the dry, bare farmland around it — a surface cool island. Shastri et al. (2017) report the same seasonal pattern for many Indian cities. The independent MODIS sensor shows no daytime heat island either (urban minus rural +0.0 °C at 1 km; Landsat at 1 km: –2.1 °C). Of the 23.82 km<sup>2</sup> of hotspots, 22.31 km<sup>2</sup> lie on rural land, 0.75 km<sup>2</sup> on peri-urban land and only 0.76 km<sup>2</sup> in urban cells.

Landsat and MODIS agree in rank (Spearman  $\rho$  0.55, R<sup>2</sup> 0.33 at 1 km), but Landsat reads 4.9 °C warmer on average (RMSE 5.1 °C). The two differ in overpass sampling, algorithm and resolution, so we compare contrasts, not absolute temperatures.

Between 2013 and 2024 urban cells became 0.31 °C cooler relative to rural land. New development cooled most (healthy growth –1.30 °C, ghost growth –1.14 °C). The 2013 composite comes from only 5 dates, so this change is indicative.

**Consequence.** Review 2's "+1.10 °C mean urban heat-island intensity" was the mean over the warmer-than-rural cells only. The heat-vulnerability layer, which weights heat by the number of residents, remains the useful planning output.

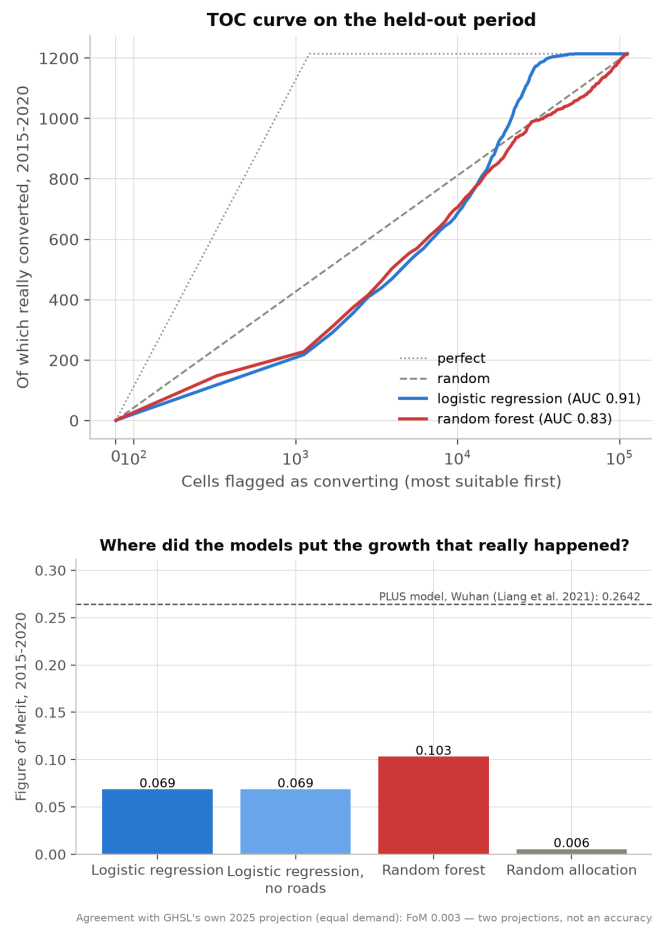
5.8 Green cover

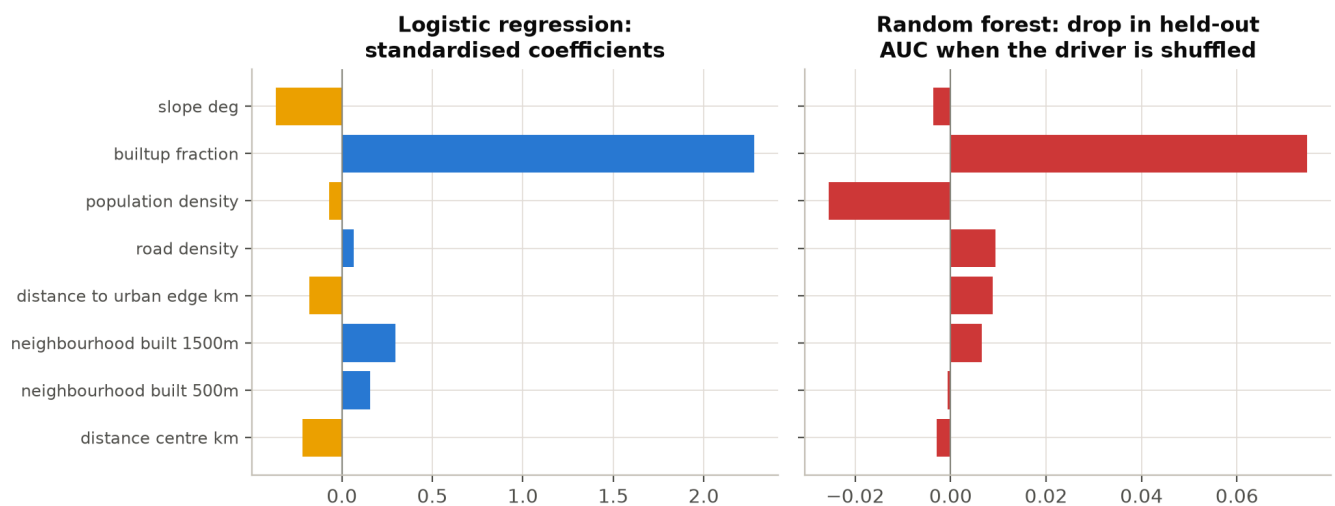
Comparing NDVI for October–March 2018-19 and 2024-25: 6.37 km<sup>2</sup> of green cover was lost and 51.31 km<sup>2</sup> gained, and green cover went from 71.0% to 77.9% of the study area — mostly the cropping calendar. Green cover lost **to built-up**, using Dynamic World gain over the same years, is **0.67 km<sup>2</sup>** (10.5% of all loss). Review 2 reported 0.05 km<sup>2</sup> using mismatched periods. Dynamic World shows 28.05 km<sup>2</sup> of built-up gain in 2018–2024.

5.9 Growth model and projections

| Model                         | AUC, training period | AUC, held-out period | Figure of Merit | × random     |
|-------------------------------|----------------------|----------------------|-----------------|--------------|
| Logistic regression           | 0.990                | 0.910                | 0.069           | 12.5×        |
| Logistic regression, no roads | 0.990                | 0.910                | 0.069           | 12.5×        |
| <b>Random forest</b>          | 0.993 (out-of-bag)   | 0.834                | <b>0.103</b>    | <b>18.7×</b> |
| Random allocation             | —                    | 0.500                | 0.0055          | 1×           |

Training period 2010–2015: 113,082 eligible (non-urban) cells, of which 1,382 became urban (1.22%). Held-out period 2015–2020: 1,214 cells became urban.

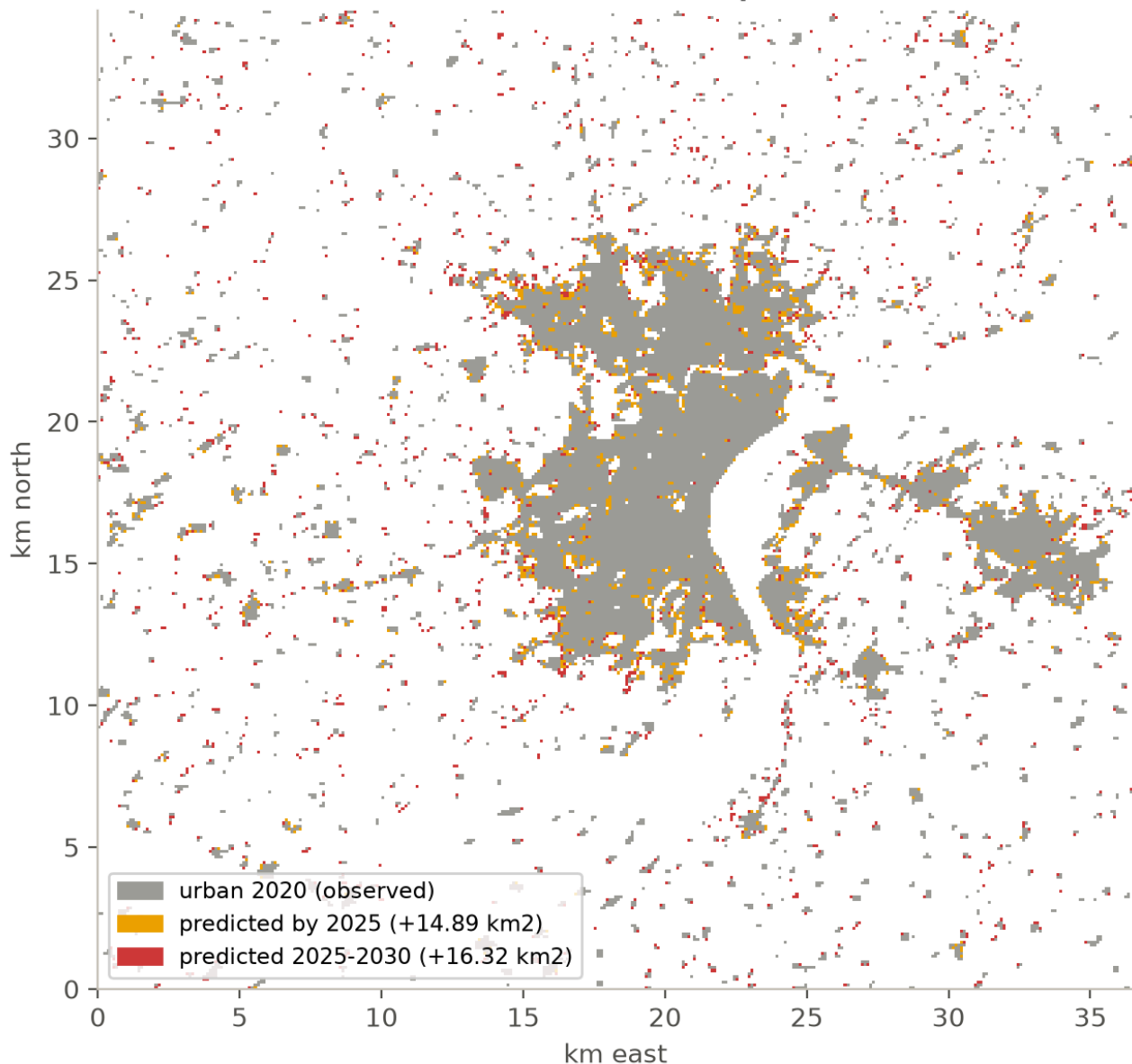




### Reading the scores.

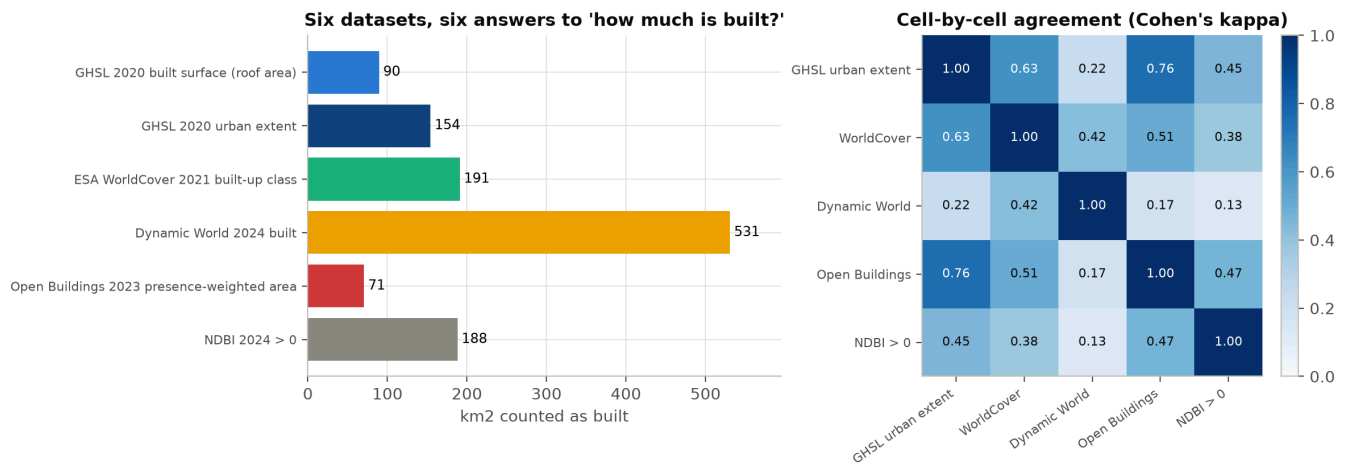
- The random forest puts more of the real growth in the right place — 227 correct cells out of 1,214, against 156 — so its Figure of Merit is higher. It ranks the bulk of the cells less well (AUC 0.83 against 0.91). The TOC curve shows why: the forest is sharper at the top of its ranking, which is the part the allocation step uses.
- The training AUC of 0.99 overstates skill. The held-out AUC is the honest figure; Review 2 quoted only the training value.
- The existing built-up fraction dominates (logistic coefficient 2.28; random-forest importance 0.075), followed by neighbourhood density and distance to the urban edge.
- Removing roads changes nothing (Figure of Merit 0.0687 both ways), so the present-day road snapshot is not leaking into the result.
- Slope: the median is 1.95° on a flat plain, which is mostly noise in the SRTM elevation model. Its negative logistic coefficient (−0.37) probably reflects river banks, and the random forest gives it no importance.
- Population density *lowers* the forest's held-out AUC when kept (importance −0.026), so it is a candidate to drop.
- For scale: PLUS in Wuhan reached a Figure of Merit of 0.264 (Liang et al. 2021). It is a different city, set of land classes and period — a reference level, not a like-for-like comparison.

### Predicted new urban land (random forest) — a prediction, not an observation



**Projection (a prediction, not an observation).** Continuing 2010–2020's growth of urban extent (1.86% a year) and placing it with the random forest: **+14.89 km<sup>2</sup> by 2025 and +31.21 km<sup>2</sup> by 2030**, measured from 2020. GHSL's own projection adds only 3.77 km<sup>2</sup> by 2025; given the same amount of growth, our map and GHSL's agree at a Figure of Merit of 0.003. The two projections disagree on both amount and location — neither 2025 map is an observation. The map also shows the second step scattering some growth around villages; the neighbourhood weight is a tuning item for Review 4.

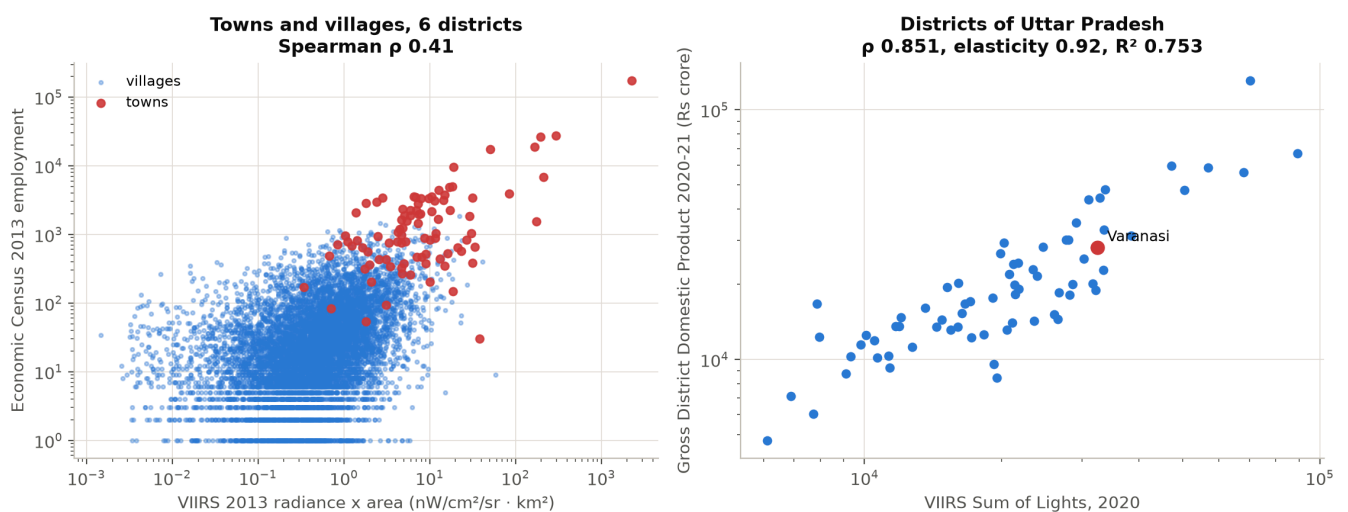
## 5.10 How much is built? Six datasets



| Dataset                                          | Built, km <sup>2</sup> | Agreement with GHSL urban extent (Cohen's $\kappa$ ) |
|--------------------------------------------------|------------------------|------------------------------------------------------|
| GHSL 2020 built surface (roof area)              | 89.7                   | —                                                    |
| GHSL 2020 urban extent (cells $\geq 20\%$ built) | 154.4                  | 1                                                    |
| ESA WorldCover 2021 built-up class               | 191.3                  | 0.63                                                 |
| Dynamic World 2024, most frequent label          | 531.0                  | 0.22                                                 |
| Open Buildings 2023, presence-weighted           | 70.7                   | <b>0.76</b>                                          |
| NDBI 2024 above zero                             | 188.3                  | 0.45                                                 |

The datasets measure different things — roof area, a land-cover class, a spectral index — so they are not expected to match. Cohen's kappa measures agreement beyond chance (1 = identical maps, 0 = no better than chance). Open Buildings agrees best with GHSL. WorldCover counts 132 km<sup>2</sup> as built that GHSL does not: village clusters, roads and yards. Dynamic World's built class spreads into 479 km<sup>2</sup> of villages and dry-season fields, so here it is used only for water and for the direction of change, never as a built-up map.

## 5.11 Does night-time light measure economic activity here?



**Towns and villages, 2013** (six districts: Varanasi, Chandauli, Jaunpur, Ghazipur, Mirzapur, Sant Ravidas Nagar; 10,143 places including 91 towns):

| Pair                                           | Spearman $\rho$ |
|------------------------------------------------|-----------------|
| Night-time light vs Economic Census employment | 0.41            |
| Night-time light vs population                 | 0.47            |
| Population vs employment                       | 0.69            |
| Light per person vs jobs per person            | 0.12            |

**Districts of Uttar Pradesh** (71 districts at 2011 boundaries):

| Pair                                             | Spearman $\rho$ | Elasticity (log-log slope) | R <sup>2</sup> |
|--------------------------------------------------|-----------------|----------------------------|----------------|
| Sum of Lights 2020 vs GDDP 2020-21 (revised)     | 0.85            | 0.92 ± 0.06                | 0.75           |
| Sum of Lights 2021 vs GDDP 2021-22 (tentative)   | 0.86            | 0.96 ± 0.06                | 0.77           |
| Per person, 2020-21                              | 0.58            | 0.76                       | 0.49           |
| Sum of Lights 2013 vs Economic Census employment | 0.70            | 0.68                       | 0.59           |

Varanasi's GDDP in 2020-21 (₹28,006 crore) is 95% of what its lights predict.

**Meaning.** Night-time light is a sound proxy for economic activity at district scale — the scale of most published evidence (Henderson et al. 2012). At village scale it mostly measures settlement size, and population predicts jobs better than light does. This is why the ghost-growth screen combines light with points of interest and population, reports at 500 m, and only claims zones of at least 0.25 km<sup>2</sup>, not single cells.

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## 6. Corrections log

"Before" is what Review 2 showed; "after" is this report.

| #   | What was wrong                                                                                                                           | Fix                                                                                     | Before                                                                                                                                   | After                                                                                                                |
|-----|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| D1  | GHSL 2025, a model projection, was the "current" epoch everywhere<br>( <code>config/varanasi.yaml</code> ,<br><code>pipeline.py</code> ) | observed epochs only;<br><code>check_epochs()</code> refuses<br>projections; 2 tests    | +23.47 km <sup>2</sup><br>built,<br>1.40%/yr,<br>13.45 km <sup>2</sup><br>new urban,<br>48.8%<br>leapfrog, 0.35<br>km <sup>2</sup> ghost | +17.25 km <sup>2</sup> ,<br>1.86%/yr,<br>10.66 km <sup>2</sup> ,<br>44.3%<br>leapfrog, 1.44<br>km <sup>2</sup> ghost |
| D2  | Water was not excluded from the heat-island rural reference, though code comments and the dashboard said it was                          | Dynamic World water and recently-built land removed                                     | reference<br>41.12 °C,<br>hotspots<br>26.57 km <sup>2</sup>                                                                              | 41.42 °C, 23.82<br>km <sup>2</sup>                                                                                   |
| D3  | "Mean urban intensity" was the mean over all warmer-than-rural cells                                                                     | mean over urban cells; old figure kept as<br><code>mean_positive_intensity_c</code>     | +1.10 °C                                                                                                                                 | <b>−1.66 °C</b>                                                                                                      |
| D4  | <i>Emerging</i> vs <i>ghost</i> used the raw sign of the light trend, with no significance test, while the whole city brightened         | relative-to-city trend with $p \leq 0.10$ ; four rules reported; hold-out test          | 95% filling up                                                                                                                           | 74%, hold-out<br>$p = 0.010$                                                                                         |
| D5  | Allocation dropped <code>demand mod 8</code> cells                                                                                       | remainder placed in the last iteration; test                                            | 1,208 of 1,214 placed                                                                                                                    | 1,214 of 1,214                                                                                                       |
| D6  | Only the training (in-sample) AUC was reported                                                                                           | held-out AUC added                                                                      | 0.9901                                                                                                                                   | 0.910 (LR),<br>0.834 (RF)                                                                                            |
| D7  | Slope was listed as a driver in the documents but never computed                                                                         | slope from <code>USGS/SRTMGL1_003</code>                                                | —                                                                                                                                        | coefficient<br>−0.37; no RF importance                                                                               |
| D8  | Present-day roads used to explain 2010–2015 growth                                                                                       | model refitted without roads                                                            | —                                                                                                                                        | no change<br>(FoM 0.0687 both)                                                                                       |
| D9  | Only a +10-year projection; the approved design promises +5 and +10                                                                      | two five-year steps                                                                     | +31.21 km <sup>2</sup><br>by 2030                                                                                                        | +14.89 km <sup>2</sup> by<br>2025, +31.21<br>km <sup>2</sup> by 2030                                                 |
| D10 | Landsat mask used QA bits 3–4 only; no depth check; the date record counted Landsat 8 only                                               | bits 1–4; $\geq 4$ dates; Landsat 8 and 9 counted                                       | "10 scenes on<br>5 days"                                                                                                                 | 22 scenes on<br>11 days                                                                                              |
| D11 | Green loss 2018–2024 intersected with built-up gain 2010–2025                                                                            | Dynamic World gain over the same years                                                  | 0.05 km <sup>2</sup><br>(0.8%)                                                                                                           | 0.67 km <sup>2</sup><br>(10.5%)                                                                                      |
| D12 | Dynamic World, Open Buildings, NDBI and LST 2013 were downloaded but unused; SRTM, MODIS and WorldCover were configured but unused       | all used (\$5.6, \$5.7, \$5.9, \$5.10)                                                  | —                                                                                                                                        | —                                                                                                                    |
| D13 | Dashboard headline showed "Built-up surface 2025" unlabelled; no growth-model output                                                     | observed values labelled; projection captioned; model layers and a Validation tab added | —                                                                                                                                        | —                                                                                                                    |

| #   | What was wrong                                                                                                                                                                                                | Fix                                                                                                         | Before | After                                           |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|-------------------------------------------------|
| D14 | Documents contradicted the code: LST "emissivity correction" (none is applied — USGS already does it), water exclusion, slope, 2025 as observed                                                               | <code>gee.py</code> docstring fixed; correction notices on the Review 2 documents; method documents updated | —      | —                                               |
| D15 | "Built-up grew 2.3× faster than population" rested on GHS-POP alone. Our own planning audit also mis-stated WorldPop's error (–12.0% for Varanasi district) by summing its per-pixel counts on the wrong grid | both datasets checked against the Census on their own grids                                                 | 2.3×   | 1.3×–2.3×; WorldPop +2.1% for Varanasi district |

## 7. Review 2 panel comments and action taken

Project Guidelines §2: review comments must be recorded and the action taken presented at the following review (1 mark at Review 3). **To be completed by the team before 15 September from the Review 2 panel's comments.**

| # | Panel comment (as recorded at Review 2) | Action taken | Where to see it |
|---|-----------------------------------------|--------------|-----------------|
| 1 |                                         |              |                 |
| 2 |                                         |              |                 |
| 3 |                                         |              |                 |
| 4 |                                         |              |                 |

## 8. Limitations

- **No ground truth for the ghost flag.** How often a flagged zone is truly vacant is unmeasured. The hold-out test (§5.5) and the Open Buildings check (§5.6) are indirect evidence, and the zone cards are for a human reader.
- **Scale of night-time light.** VIIRS pixels are about 463 m across; light is a weak proxy at village scale (§5.11).
- **Population.** GHS-POP over-counts the city core by 63% at the 2011 town boundary; the activity index still uses it, with a weight of 0.20.
- **Heat.** Daytime only; the 2013 composite has 5 dates; Landsat and MODIS differ by 4.9 °C in absolute terms.
- **Dynamic World's built class** is unreliable in this landscape (§5.10).
- **Projection.** Demand assumes the 2010–2020 rate continues; roads and population are held at 2020 values; the second step scatters some growth.
- **OpenStreetMap** points of interest are thin on the periphery and are a present-day snapshot.
- The **2013 Economic Census** is the latest one available to us at town and village level. District comparisons use 2011 boundaries: Shamli, Hapur, Sambhal and Amethi, created after the Census, are

added back to the districts they came from.

## 9. Plan for Review 4

| Item                     | Detail                                                                                                                                                              |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| City search              | Nominatim geocoding → circle → GHSL from Earth Engine ( <code>JRC/GHSL/P2023A/GHS_BUILT_S</code> , checked against the bulk tile) → automatic UTM zone              |
| Labels for flagged zones | image interpretation of a sample of flagged and unflagged cells on historical high-resolution imagery, to measure the precision of the ghost flag                   |
| Activity index           | a WorldPop-based variant; drop the population driver from the growth model if the held-out result confirms it                                                       |
| Planning data            | ward-level reporting (Census ward tables and ward boundaries, after a licence check); VDA Master Plan 2031 overlay; Bhuvan NUIS/AMRUT land use; UP RERA cross-check |
| Growth model             | tune the neighbourhood weight; report a range of demand scenarios                                                                                                   |
| Publication              | a paper submission for Review 4 item 9, built on §5.5 and §5.11                                                                                                     |
| Report                   | Chapters 1–4 near-final; individual reports                                                                                                                         |

## 10. Contributions and AI acknowledgement

Who owns each work package is recorded in `../CONTRIBUTIONS.md`. Each owner re-runs, explains and presents their own part.

**AI acknowledgement (Project Guidelines §2).** An AI assistant (Claude, by Anthropic) was used for code, analysis scripts and documentation in this project, including the work reported here. Every result comes from scripts in this repository and can be reproduced with `run_review3.bat`; the team is responsible for understanding, checking and presenting each part.

## 11. References

### Papers

- Asher, S., Lunt, T., Matsuura, R. and Novosad, P. (2021). Development research at high geographic resolution: an analysis of night-lights, firms, and poverty in India using the SHRUG open data platform. *The World Bank Economic Review* 35(4).
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- Pontius, R. G. Jr. et al. (2008). Comparing the input, output, and validation maps for several models of land change. *The Annals of Regional Science* 42(1), 11–37.
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- Shastri, H., Barik, B., Ghosh, S., Venkataraman, C. and Sadavarte, P. (2017). Flip flop of day-night and summer-winter surface urban heat island intensity in India. *Scientific Reports* 7, 40178.

The twenty papers reviewed at Review 2 are summarised in [PAPER\\_SUMMARIES.md](#).

**Datasets** — as listed in §3, with official identifiers. SHRUG data are used under CC BY-NC-SA 4.0 and cited as Asher et al. (2021).